

Doctoral Program Research Proposal

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Research Plan Proposed Topic/Title of Research: A Longitudinal Cohort Framework for Early Detection of Type 2 Diabetes Risk Using Integrated Lifestyle and Genomic Data. Background: Type 2 diabetes mellitus (T2DM) is a chronic metabolic disease whose global prevalence has increased rapidly over the past decades. According to the World Health Organization, diabetes is now one of the leading causes of mortality and morbidity worldwide, posing significant challenges to healthcare systems, particularly in developing countries. The increasing burden of T2DM highlights the urgent need for more effective prevention and early detection strategies. The development of T2DM is strongly influenced by modifiable lifestyle factors, including unhealthy dietary habits, physical inactivity, smoking, and poor sleep patterns. Studies published in The Lancet have demonstrated that modern lifestyle transitions are a major contributor to the rising incidence of metabolic disorders, especially among younger populations (Zheng et al., 2018). However, lifestyle factors alone cannot fully explain the variation in disease susceptibility across individuals. Genetic predisposition has been identified as a key determinant of individual risk. Research in Nature Reviews Genetics shows that specific genetic variants significantly influence insulin resistance and glucose metabolism (Mahajan et al., 2018). Furthermore, integrating genetic data with other health-related data has been shown to improve the accuracy of disease risk prediction models (Khera et al., 2018). Despite these advancements, many existing studies analyze lifestyle and genetic factors separately, resulting in fragmented insights. Such approaches fail to capture the complex interactions between environmental exposures and biological mechanisms. Therefore, there is a growing need for integrative frameworks that combine multiple data types to provide a more comprehensive understanding of disease risk. In addition, a large proportion of previous studies rely on cross-sectional designs, which only provide a static snapshot of an individual’s condition. This limits the ability to understand how risk evolves over time. In contrast, longitudinal cohort approaches allow continuous monitoring of individuals, enabling more accurate assessment of disease			

progression and temporal risk dynamics (Caruana et al., 2015).

Based on these gaps, this study proposes the development of a longitudinal cohort framework that integrates lifestyle and genomic data for early detection of type 2 diabetes risk. The main research question addressed in this study is:

How can the integration of lifestyle and genomic data within a longitudinal cohort framework improve early prediction of type 2 diabetes risk?

Methodology:

This study adopts a longitudinal cohort design to develop a predictive framework for early detection of type 2 diabetes risk through the integration of lifestyle and genomic data.

1. Study Population and Data Source

The study will involve a cohort of individuals aged 18–60 years who are initially free from type 2 diabetes. Participants will be recruited from publicly available cohort datasets (e.g., UK Biobank or similar open-access health databases) or simulated datasets if real genomic data access is limited.

The inclusion criteria include:

- No prior diagnosis of diabetes at baseline
- Availability of lifestyle-related data
- Availability of genetic or proxy genomic data

The expected sample size ranges from 500 to 5,000 individuals to ensure sufficient statistical power for analysis.

2. Data Collection and Variables

Two main categories of data will be used:

a. Lifestyle Variables

- Dietary patterns
- Physical activity level
- Smoking status
- Sleep duration and quality

b. Genomic Variables

- Genetic risk scores (polygenic risk score, if available)
- Selected genetic markers associated with T2DM

c. Outcome Variable

- Development of type 2 diabetes (binary outcome: yes/no)
- Proxy indicators such as fasting glucose levels or BMI changes

3. Longitudinal Data Structure

Data will be structured in a time-series format, where each individual has repeated observations over multiple time points (e.g., monthly or yearly intervals). This allows tracking of changes in lifestyle patterns and risk indicators over time.

This time-based structure enables:

- Monitoring progression of risk
- Comparing expected vs observed health trajectories
- Identifying early deviations indicating increased risk

4. Data Processing and Integration

Data preprocessing will include:

- Handling missing values using imputation techniques
- Normalization of continuous variables
- Encoding categorical variables

Lifestyle and genomic data will be integrated into a unified dataset to enable joint analysis. A composite risk profile will be generated for each individual at each time point.

5. Predictive Modeling Approach

To predict the risk of type 2 diabetes, the study will employ:

- Logistic Regression (baseline model)
- Random Forest or Gradient Boosting (for improved prediction accuracy)

The model will be trained to estimate the probability of developing diabetes based on both lifestyle and genomic inputs.

Model performance will be evaluated using:

- Accuracy
- ROC-AUC
- Precision and recall

6. Longitudinal Monitoring Framework

A key component of this study is the development of a time-based monitoring framework that tracks individual risk progression over time.

This framework includes:

- Calculation of expected risk trajectory based on baseline characteristics
- Comparison with actual observed progression

- Identification of risk deviation (gap analysis)

Individuals will be categorized into:

- Normal risk progression
- Elevated risk
- High-risk (requiring early intervention)

7. Statistical Analysis

Descriptive and inferential statistical methods will be applied:

- Descriptive statistics for baseline characteristics
- Correlation analysis between variables
- Regression analysis to identify significant predictors
- Longitudinal trend analysis

Statistical significance will be evaluated at a 95% confidence level.

Work Plan:

This research is planned to be completed over a period of approximately 12 months. The study is structured into several phases, from data acquisition to model development and evaluation, ensuring a systematic and feasible workflow.

Phase 1: Literature Review and Research Design (Month 1–2)

- Conduct an in-depth review of existing literature on type 2 diabetes risk factors, lifestyle influences, and genomic data integration
- Identify research gaps and refine the research framework
- Define variables, data structure, and modeling approach

Phase 2: Data Acquisition and Preparation (Month 3–4)

- Obtain datasets from publicly available cohort databases (e.g., UK Biobank or similar sources)
- Extract relevant variables (lifestyle, genomic, and outcome data)
- Perform data cleaning, including handling missing values and standardization

Phase 3: Data Integration and Structuring (Month 5–6)

- Organize data into a longitudinal format with repeated observations per individual
- Integrate lifestyle and genomic data into a unified dataset
- Construct individual risk profiles across time

Phase 4: Model Development (Month 7–9)

- Develop baseline predictive models using logistic regression
- Implement advanced models such as Random Forest or Gradient Boosting
- Train models using integrated dataset

Phase 5: Model Evaluation and Validation (Month 10)

- Evaluate model performance using metrics such as accuracy, ROC-AUC, precision, and recall
- Compare performance between baseline and advanced models
- Validate model robustness

Phase 6: Development of Longitudinal Monitoring Framework (Month 11)

- Design a time-based monitoring system to track individual risk progression
- Implement expected vs observed risk comparison
- Identify patterns of early risk escalation

Phase 7: Interpretation and Reporting (Month 12)

- Interpret findings and identify key risk factors
- Prepare research report and documentation
- Summarize implications for early detection and prevention strategies

Expected Outcomes and Value:

This study is expected to produce both methodological and practical contributions in the field of population health and disease risk prediction.

1. Development of an Integrated Predictive Framework

The primary outcome of this research is the development of a longitudinal cohort framework that integrates lifestyle and genomic data for early detection of type 2 diabetes risk. This framework is expected to provide a more comprehensive approach compared to traditional models that rely on single data sources.

2. Improved Early Detection of Disease Risk

By combining lifestyle and genetic information, the proposed model is expected to improve the accuracy of early risk prediction. This enables identification of high-risk individuals before clinical symptoms appear, supporting preventive interventions.

3. Understanding of Risk Progression Over Time

The longitudinal design allows analysis of how risk evolves over time, providing insights into dynamic changes in individual health status. This contributes to a better understanding of disease development rather than static risk assessment.

4. Contribution to Data-Driven Preventive Healthcare

The results of this study can support the development of data-driven strategies in preventive healthcare. The framework may assist healthcare providers in designing targeted interventions based on individual risk profiles.

5. Scalable Monitoring Approach for Population Health

The proposed framework has the potential to be adapted into scalable monitoring systems for large populations. It can be applied in future research or integrated into digital health platforms for continuous health monitoring.

6. Academic Contribution

This study contributes to the growing body of research on integrative approaches in disease prediction by combining longitudinal analysis with multi-source data integration. It also highlights the importance of moving beyond cross-sectional studies toward dynamic risk modeling.

References:

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